Harder Geometrical Problems — problem 12.3.2.78

A. Engel — Problem-Solving Strategies · 1998

M is a tiny Anchurian island whose territorial waters extend one mile. At night a powerful searchlight rotates slowly counterclockwise about M, illuminating the territorial waters. At B (distance 1 mile) there is a Sikinian boat whose mission it is to reach M undetected. The boat has maximum speed b. At a distance of one mile from M, the light beam of the searchlight has speed s.

(a) Suppose $k = s/b = 8$. Show that the boat can fulfill its mission.

(b) Suppose $k = s/b < 2\pi + 1$. Show that the boat can fulfill its mission.

(c) Find the smallest k for which the boat can fulfill its mission.